

IMAGE To GARMENT

Prerana Achar, Siddharth Pandey

Motivation

- 3D Generative models can generate messy topology
- Cannot be edited/sized easily
- No semantic interpretability
- Recent VLM based methods are computationally expensive



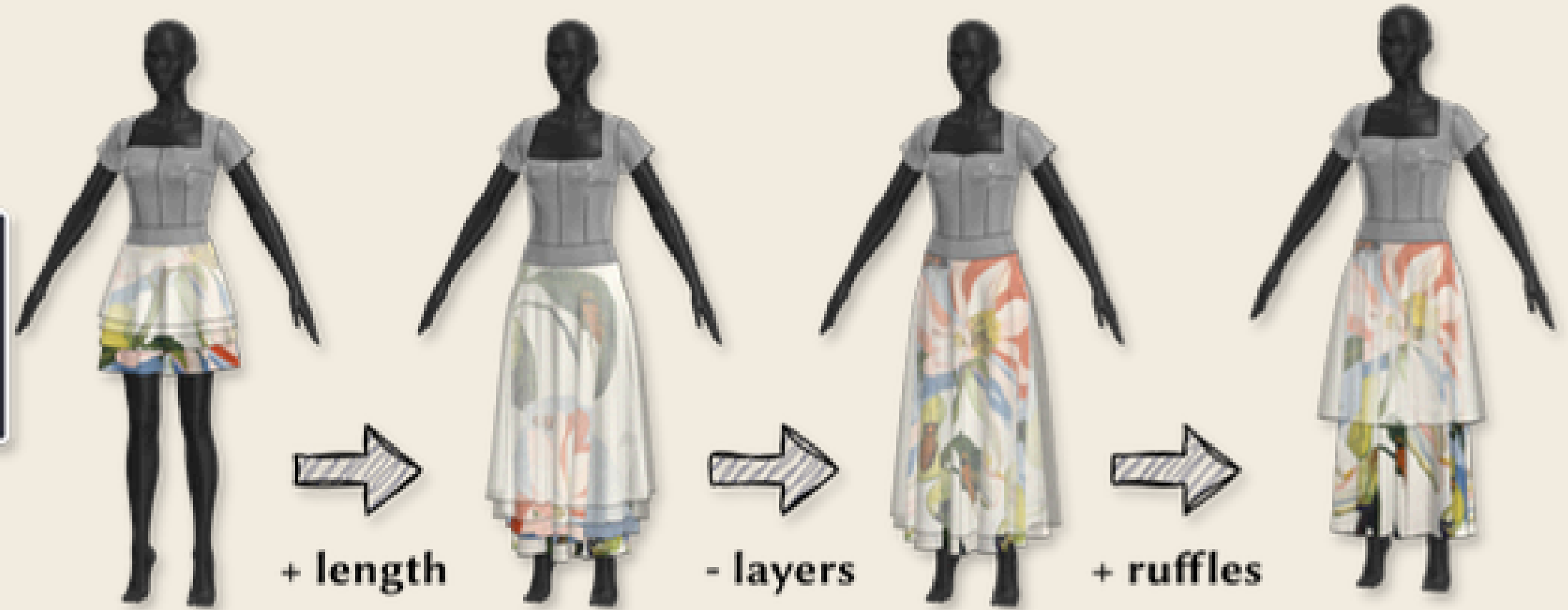
Clean quad topology
(consistent, even edge flow)



Messy topology
(hard to edit, UV, or simulate)

GarmentCode

- Parameterised
- Editable, sizable
- Can be understood by an LLM



(a) Generated Code

(b) Default Design

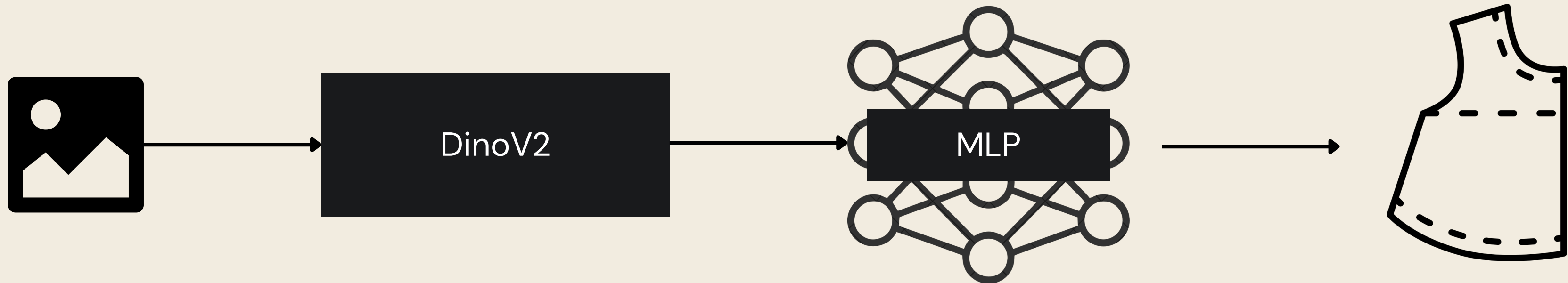
(c) Skirt **Length**

(d) Skirt **Layers**

(e) Layer **Differences**

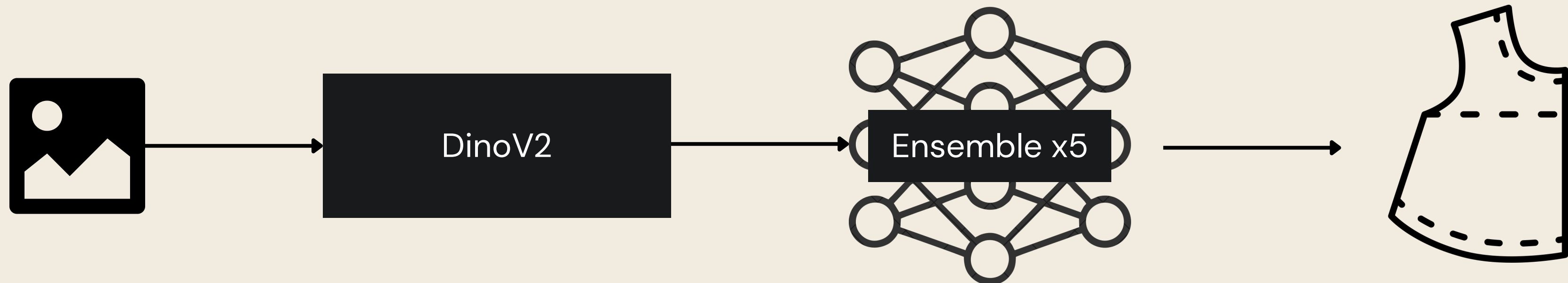
Baseline

- Simple MLP Model
- Losses used: Categorical, Numerical, Ordinal



Ensemble Model

- Extra Losses used: Pose consistency, route consistency



Dataset - ChatGarment



Dataset

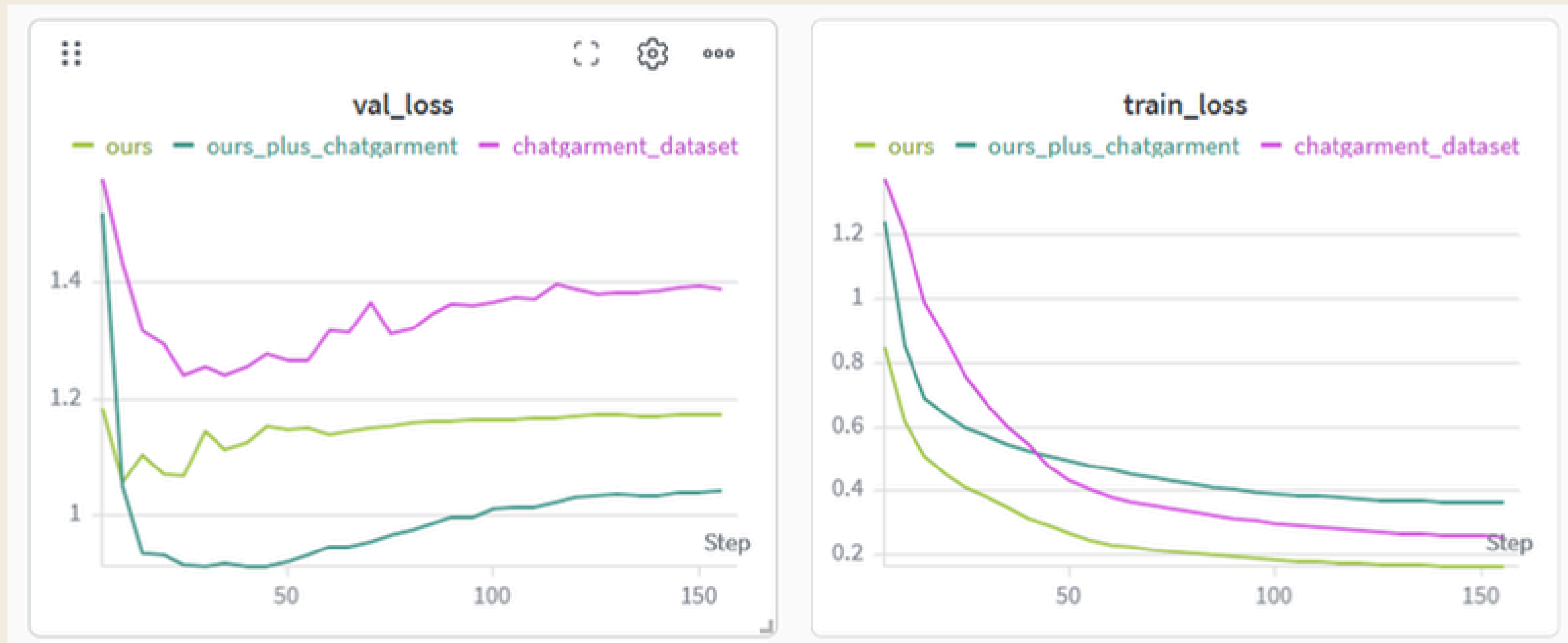


Results - Qualitative

Model	Train loss	Val loss	L2 Loss	Eval time	Time / image	Trainable Parameters
Two-head MLP	1.0932	0.9813	0.07409	1359 s	13.32 s	592,255
Multi-MLP	1.1441	0.9826	0.08423	1377 s	13.50 s	16,213,582
Three-head MLP	1.1071	0.9739	0.0852	1505 s	14.75 s	410,815
Ensemble fixed-v2	0.4892	0.6763	0.08655	1701 s	16.68 s	3,925,425
ChatGarment	—	—	0.28115	2633 s	25.81 s	312,799,308

Results - Quantitative

- The validation set contains some in the wild images
- ChatGarment is not very good but using only our images overfits
- Using both decreases the val loss



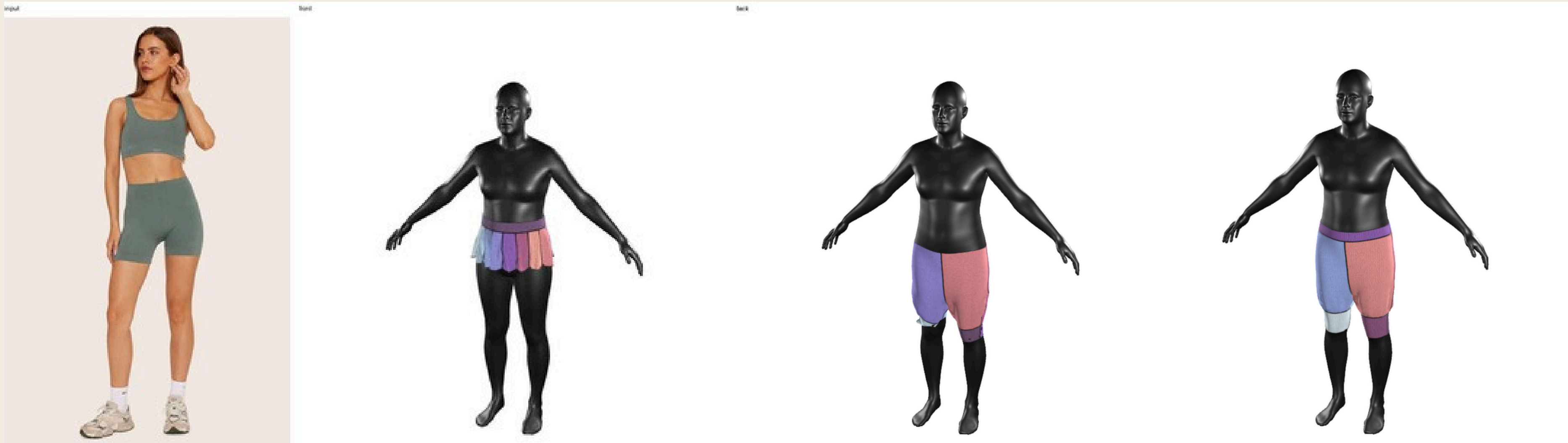
Results

Garment Reconstruction

ChatGarment

Ours (MLP)

Ours (Ensemble)



Results

Garment Reconstruction

ChatGarment

Ours (MLP)

Ours (Ensemble)



Results

Garment Reconstruction

ChatGarment

Ours (MLP)

Ours (Ensemble)



Results

Garment Reconstruction

ChatGarment

Ours (MLP)

Ours (Ensemble)



Results

ChatGarment



Ours (MLP)

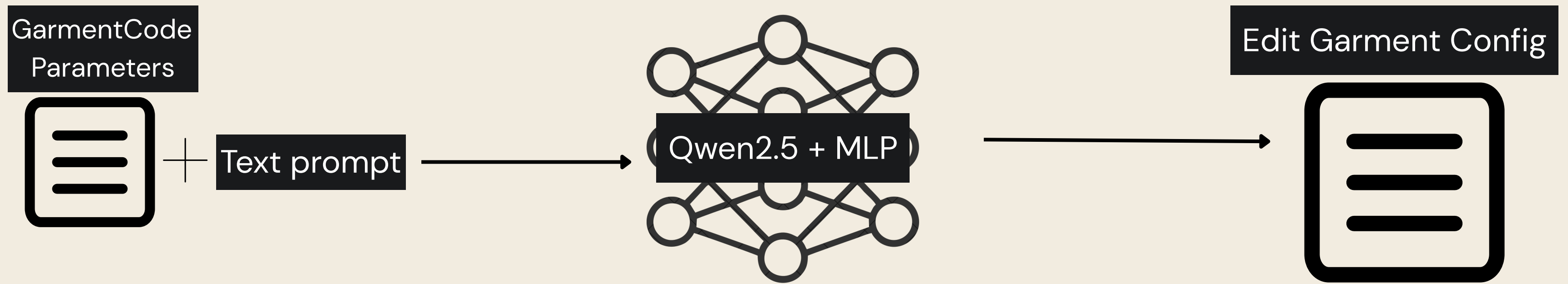


Ours (Ensemble)



Editing LLM

- Fine-tuned LLM which takes in GarmentCode parameters for a garment along with a text input to output edited garment configurations.
- Regresses over hidden activations to output continuous values.
- Only 30M trainable parameters



Results

Editing Garments

HOODIE · evaluation:5



input GarmentCode

TEXT EDIT

SLEEVE

- No visible cuffs
- Smooth arm openings
- Straight cut at shoulders
- Sleeveless design
- Wide armholes



ours · Qwen3B per-token



ChatGarment



ground truth

Results

Editing Garments

DRESS · training-edit:16-dress

TEXT EDIT

GARMENT BODY

- A-line silhouette
- floor-length
- form-fitting top
- flared skirt



input GarmentCode



ours · Qwen3B per-token



ChatGarment

Results

Editing Garments

test record 0

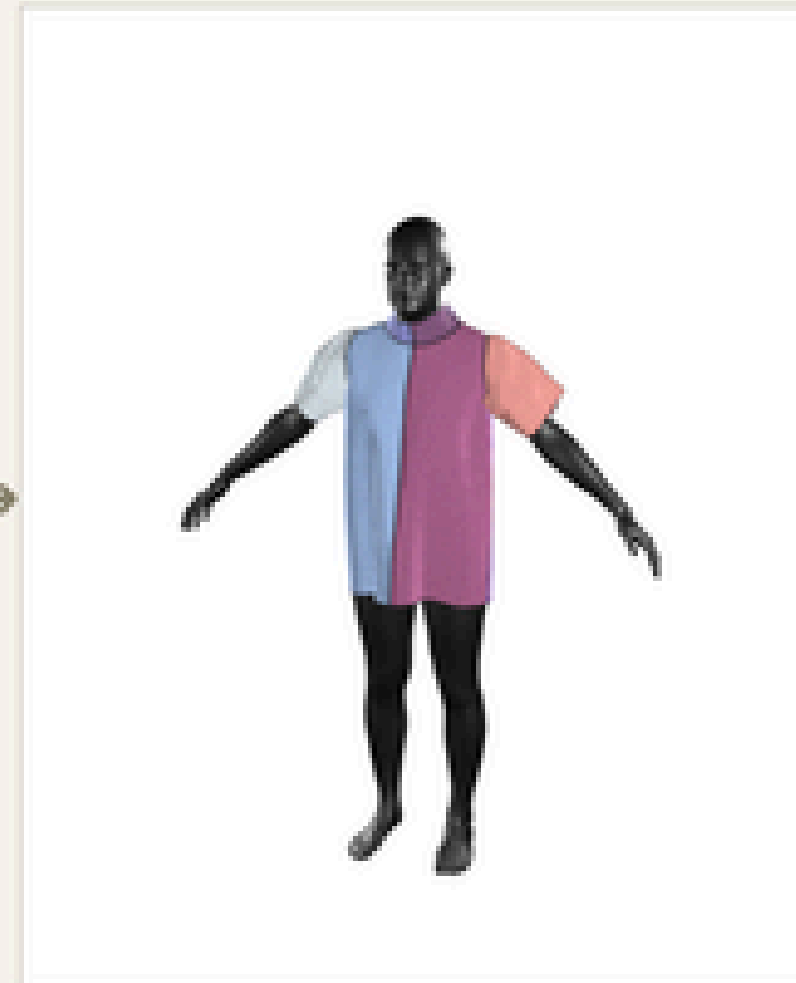


initial

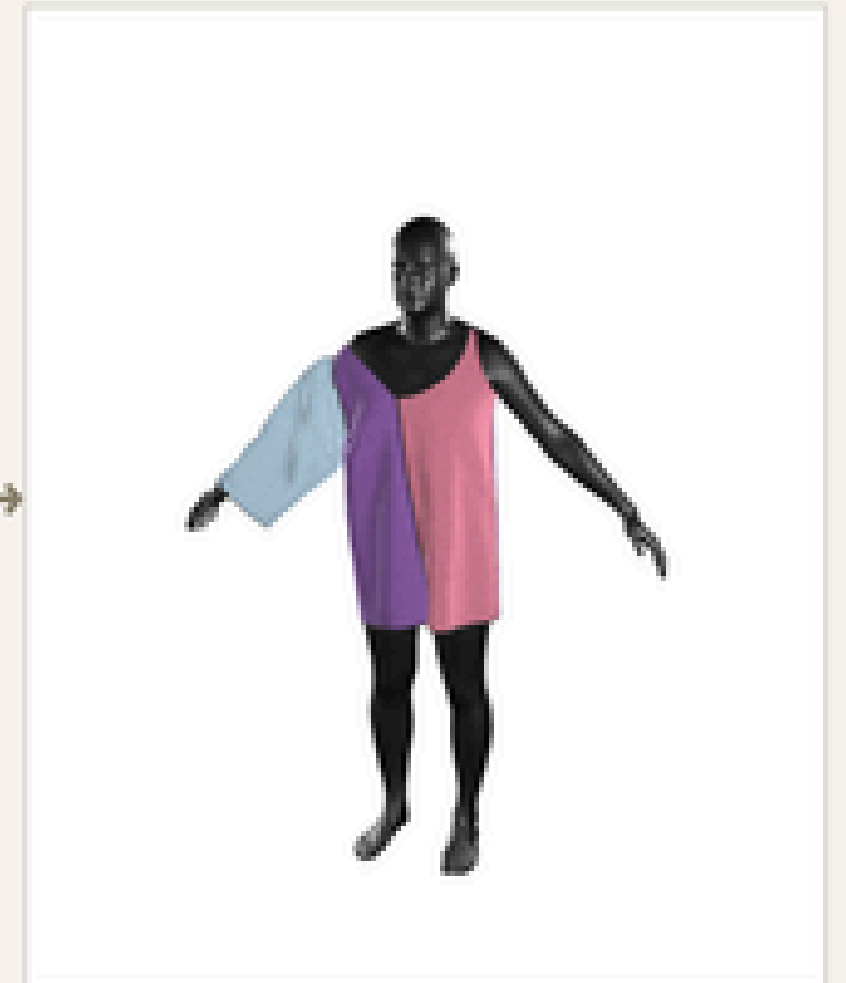
TEXT EDIT

SLEEVE

- Straight cut
- Wide opening
- Short length
- Dropped shoulder
- Loose fit



checkpoint prediction



ChatGarment output

Future Work

- Use GNN based remesher/enhancer to edit the mesh to fit 3D generated shapes
- Use more data

Thank You!